Research statement

Gene fading and gene birth after genome merger in the allopolyploid *Capsella bursa-pastoris*
Genomes are not fixed sets of genes. Over time, some genes are lost and new ones arise, sometimes from DNA that never coded for a protein. These changes shape how species adapt, yet it is still unclear which genes are lost first and how new genes begin. Polyploid plants, which carry two complete genomes after an ancestral merger, offer a natural experiment: every gene starts with a spare copy, so the fading of old genes and the birth of new ones can be studied within a single species. My research asks what decides which genes a genome gives up and what it builds anew.

Past research

My doctoral research at IISER Bhopal studied the lifecycle of genes, from birth to loss, using large public genome datasets.

Where new genes come from. New genes can arise from DNA stretches that lack "stop" signals and could therefore be read into proteins. Mapping these stretches across seven complete primate genomes showed that the longest ones already look gene-like, and that some existing genes could extend into neighbouring DNA (preprint).
How genes are lost. Tracing the immune gene *APOBEC1* across 81 bird genomes showed that it was lost many times independently, mostly in lineages exposed to fewer retroviruses (*Immunogenetics*, 2026). Combined with earlier work [1], this suggests that the gene began as an antiviral DNA editor: a useful defence that also risked damaging the host's own DNA. Birds kept this role but shed the gene when viral pressure fell. In mammals, a newer gene family took over viral defence, freeing *APOBEC1* to evolve into a precise RNA editor that helps process dietary fat. This work also required separating true losses from gaps in genome assemblies.
When a broken gene still works. The cattle gene *GPRC6A* looks broken in the genome, yet the receptor is active. Five mechanisms that could preserve its function were evaluated *in silico* using public proteomic, transcriptomic and genomic data. The results showed that cells may still produce usable products from it, so a damaged sequence does not always mean a lost function (*Cureus J. Agric. Food Sci.*, 2026).

These projects compared species. The next step is to follow gene fate within populations and in genomes with mixed ancestry, which is the focus of the proposed research.

Proposed research

Background. *Capsella bursa-pastoris* (shepherd's purse) is a widespread weed that formed when two ancestral *Capsella* lineages merged their genomes [2]. It therefore carries two copies of most genes, one from each parent, and selection on each copy is expected to relax. In allopolyploid cotton, harmful mutations accumulate faster than in diploids and unequally between the two parental genomes [3]. Which copies fade first, however, is unclear. Work in *Arabidopsis* and *Capsella* offers a clue: genes expressed only under particular conditions are under weaker selection [4], whereas genes central to gene networks are more constrained [5]. These questions can now be tested, because a *C. bursa-pastoris* genome assembly that separates the two parental genomes is available, together with genome sequences from 133 individuals [6]. This genome also carries DNA introgressed from a related species [6], so its history is a network rather than a simple tree, and methods that assume a tree can misplace genes [7, 8].
Hypothesis. After genome merger, selection relaxes unevenly: spare copies of condition-specific and weakly connected genes fade first, and the same relaxed constraint allows new open reading frames to persist.

Aim 1. Which parental copy fades first?
Why it matters. Many crops and wild plants are polyploid, yet it is hard to predict which of their duplicated genes will be kept and which will decay. Finding the rules that decide this would show how a merged

genome settles into a working whole, and whether the same forces that weaken selection within a species also drive gene loss over longer time scales.
Approach. Gene pairs will be assigned to the two parental genomes using the subgenome-resolved assembly [6], comparing tree-based and network-based placement to account for introgression [7, 8]. Copies with broken reading frames or lost expression will be identified and checked against raw sequencing reads, so that assembly errors are not mistaken for loss. Apparent breaks will also be checked for rescue at the RNA level, such as RNA editing. Selection on each copy will be measured within the species using the population genomes [6], and by comparison with relatives. To measure condition-specific expression directly, RNA-seq will be generated from the host lab's North American seed collection grown under a few contrasting conditions, with public *C. grandiflora* data as a single-condition baseline for the diploid parent.
Expected results and alternatives. The aim will produce a map of which gene copies are fading in *C. bursa-pastoris* and why. Copies of condition-specific and weakly connected genes are expected to show more breaks and weaker selection. If no such pattern appears, gene dosage or parental origin may matter more; both can be tested with the same data.

Aim 2. Do new genes begin after genome merger?
Why it matters. New genes are a source of evolutionary novelty, but how a gene begins from non-coding DNA remains one of the open questions in evolution. A genome merger may be a moment when new genes are more likely to arise, and this aim tests that idea directly in a recently formed species.
Approach. Both parental genomes and those of related species will be scanned for long DNA stretches without stop signals, and sorted into those shared with relatives and those found only in *C. bursa-pastoris*. Because introgression can make an old stretch look new, the age of each will be estimated on a network rather than a single tree [7, 8]. Candidates will be tested for selection using the population genomes and established approaches [6, 9], and for activity using the new RNA-seq data and, where available, public protein data. Finding matching non-coding DNA in relatives will support a truly new origin.
Expected results and alternatives. The aim will produce a catalogue of candidate new genes, with evidence of which are active and under selection. The allopolyploid is expected to carry more of these open stretches than its diploid relatives, with most evolving neutrally and a few showing constraint. If no excess is found, genome merger does not promote new genes on this time scale, which would itself narrow where new genes arise. Together, the two aims will show, within one species, how a genome gives up old genes and begins new ones.

Host labs and timeline

Suitability of the host labs. The project combines the complementary expertise of two labs in the Ecology, Evolution, and Behavior (EEB) program at Michigan State University (MSU): the Josephs lab, which studies selection and gene expression in *Capsella* and *Arabidopsis* [4, 5, 6, 9], and the Liu lab, which develops phylogenetic methods for non-tree-like evolution [7, 8]. Together with my experience in gene-loss validation and genome-scale mapping of open reading frames, this setting supports the full project and provides training in population genetics and method benchmarking.

Timeline. Months 1–8: assignment of gene copies to parental genomes, identification of fading copies, and plant growth and RNA-seq. Months 9–14: selection and expression analyses for Aim 1; first manuscript. Months 15–22: scans, network-based dating and selection tests for Aim 2. Months 23–24: second manuscript and release of all code and data.

References. [1] Severi et al. (2011) *Mol. Biol. Evol.* [2] Douglas et al. (2015) *PNAS.* [3] Conover and Wendel (2022) *Mol. Biol. Evol.* [4] Roberts and Josephs (2023) *Genetics*. [5] Josephs et al. (2017) *Genome Biol. Evol.* [6] Wilson Brown et al. (2026) *bioRxiv*. [7] Alamin and Liu (2025) *IEEE/ACM TCBB*. [8] Gao et al. (2022) *ACM-BCB*. [9] Williamson et al. (2014) *PLoS Genet.*